

Phase-Gating Across Precision Channels: Topological Constraints on Multi-Channel Belief Update Dynamics

Alexander Sabine

Independent Researcher

Alexander@activeinference.institute

https://www.temporalgrammar.ai/human_networks.html

Abstract. Every finite system that persists must accumulate evidence, reach the limits of its current regime, and reconstruct from its own history. We propose that these three temporal operations, which we call Coherence, Rupture, and Regeneration (CRR), constitute a candidate temporal grammar whose dynamics are governed by a single geometric parameter Ω , fixed by the topology of the system’s statistical manifold. Two fundamental symmetry classes yield two thresholds without free parameters: π for bistable ($\mathbb{Z}_2$) systems, 2π for rotational ($\text{SO}(2)$) systems, with a ratio of exactly 2. The parameter Ω functions analogously to a cognitive light cone [4]: it governs both the depth of accessible history and the breadth of future possibility.

We apply CRR to a concrete problem in multi-channel AGI architecture: how do multiple precision channels coordinate their belief updates in predictive coding? Assigning $\mathbb{Z}_2$ symmetry to sensory precision and $\text{SO}(2)$ to prior precision, we test the resulting dynamics in a POMDP with Dirichlet learning. The primary finding is phase-gating: the topological thresholds produce a strongly non-uniform phase relationship between channels ($\chi^2 = 8,041$) that determines whether each update drives learning or action. This result is consistent across environment topologies, independent of the weight function, and structurally compatible with recent empirical findings on neuromodulatory timing. We present CRR as a parsimonious temporal constraint for multi-channel AGI architectures, with potential applications in alignment, continual learning, and applied Active Inference.

Keywords: Active Inference · Temporal grammar · Precision weighting · Information geometry · Phase-gating · AGI architecture

1 Introduction

The Free Energy Principle (FEP) provides a powerful account of how adaptive systems maintain their statistical boundaries by minimising variational free energy [1]. In Active Inference, this account extends to action and perception through a hierarchy of precision-weighted prediction errors [2]. Distinct precision

channels govern sensory confidence (likelihood precision), model confidence (prior/transition precision), and policy confidence. The textbook treatment specifies *what* each channel does but leaves open the temporal structure of *when* each channel updates, *how much* information it absorbs per update, and how the channels coordinate with one another. This temporal structure features in recent empirical work by Jang et al. [3] who demonstrate that the relative timing of neuromodulatory precision signals, not their magnitude, determines whether the same dopamine signal promotes reinforcement learning or movement vigour.

This paper introduces the Coherence-Rupture-Regeneration (CRR) framework as a temporal grammar for self-organising systems. CRR derives the update dynamics of bounded systems from information geometry, connecting to the Cramér-Rao bound [24,25] and the thermodynamic speed limits established by Ito and Dechant [6]. The single parameter Ω functions as an information-theoretic capacity limit: the geodesic extent of the system’s statistical manifold [12,13] determines Ω uniquely, yielding $\Omega = 1/\pi$ for systems with $\mathbb{Z}_2$ (bistable) symmetry and $\Omega = 1/2\pi$ for systems with $\text{SO}(2)$ (rotational) symmetry.

The observation that motivated the present study arose from an interactive network simulation of agents connected by Markov blankets.¹ The symmetry assignment emerged from the graph structure itself, where each node represents an internal model that traverses a continuous cycle of belief updating ($\text{SO}(2)$ dynamics), while each edge is a statistical boundary that alternates between two regimes of influence ($\mathbb{Z}_2$ dynamics). In other words, the two symmetry classes were not awkwardly imposed on the FEP’s precision architecture; they were found in its graph topology.

The question this paper investigates is as follows: if two precision channels have different symmetry classes, what does the topology enforce about their coupled dynamics? We find that the topology enforces a conservation law: both channels must process equal total precision-gain per unit time. We show in Section 4 that this conservation is a structural consequence of equal evidence delivery rather than a distinctive prediction of CRR; what the topology contributes is not the conservation principle itself, but the specific temporal decomposition and, crucially, the phase-gating dynamics that emerge from the $\pi/2\pi$ partition. We also found that adding a third channel does not automatically extend pairwise conservation, establishing an honest boundary on the claim, while highlighting wider implications for distributed versus centralised AGI systems. Indeed, these findings suggest that the temporal coordination of precision channels in AGI architectures should respect the topological constraints of their respective state spaces. On this final point, we simply ask the reader to play with the pedagogical CRR Active Inference Network simulation and reflect intuitively on their experience; the parsimony of the framework – a single geometric parameter governing temporal coordination – is perhaps best appreciated through direct engagement with the dynamics it produces.

¹ An interactive version of this simulation is available at https://www.temporalgrammar.ai/human_networks.html.

2 Related Work

2.1 From Spatial Boundaries to Temporal Dynamics

The FEP characterises the *spatial* structure of inference. Da Costa et al. [8] showed that internal states parametrise beliefs and flow as gradient descent on model evidence; Kirchhoff et al. [10] extended this to biological autonomy; the path integral formulation [9] describes how trajectories on statistical manifolds are weighted by accumulated action. What remains underspecified, however, is the *temporal* structure - when must the system reorganise, and how does it coordinate belief updates across precision channels? CRR takes a process-oriented approach, complementing the FEP’s hierarchical generative models [2] and resonating with Levin’s TAME framework [4]. The cognitive light cone concept provides a measure of the temporal and spatial scale of an agent’s goal-directedness; in CRR, Ω plays an analogous role, governing both the depth of past access and the breadth of future possibility (Fig. 2). Indeed, Tolchinsky, Levin et al. [5] have also explored how temporal depth serves as a parameter whose collapse characterises dissociative states; CRR offers a geometric account of what determines temporal depth.

2.2 Speed Limits and the Geometry of Evidence Accumulation

A system traversing a statistical manifold is subject to fundamental speed limits - the rate of state change is bounded by the Fisher information [6], and thermodynamic fluctuation timescales are universally bounded by the same geometry [7]. Accordingly, Parr, Da Costa, and Friston [23] showed that Markov-blanketed systems have both an information geometry and a stochastic thermodynamics. Nevertheless, these limits specify the maximum rate of evidence accumulation, but do not fully explain what happens *at* the limit, when the system has traversed its manifold’s full geodesic extent and the current regime of free energy minimisation is exhausted. CRR proposes that this moment is ‘rupture’, and that the geodesic extent (π for Bernoulli, 2π for S^1) determines the threshold.

2.3 The Timing Problem in Precision Weighting

Precision weighting is central to Active Inference [2], where distinct neuromodulatory channels govern prediction error confidence [16], but their temporal coordination remains open. Three recent findings render this gap more visible. Krok et al. [26] showed that dopamine and acetylcholine fluctuate spontaneously at approximately 2 Hz in the striatum with a 90-degree phase offset, even without salient stimuli. Taniguchi and Tritsch [27] found that cholinergic pause and dopamine signal coincidence varies by subregion, ruling out a simple gating model. Most directly, Jang et al. [3] demonstrated that relative timing, not magnitude, determines whether dopamine promotes learning or movement. The hypothesis that cholinergic and dopaminergic signals must be temporally

coincident for synaptic plasticity [22,18,17] further motivates the search for a principled account of inter-channel timing.

CRR provides a geometric account of temporality, where channels with different symmetry classes accumulate evidence at different rates, producing a phase relationship that emerges from topology rather than from a separately dedicated ‘top-down’ controller. This perspective is consonant with work on inter-agent coordination [20], information-geometric constraints on development [21], and the duality between the FEP and the maximum-entropy principle [11]. The present work addresses both the within and between-agent versions of this coordination problem.

3 The CRR Framework

CRR models bounded systems through three coupled temporal processes (Fig. 1), resting on five core commitments that any persistent, finite, adaptive system must satisfy. We present these commitments because they determine the temporal decomposition and phase-gating dynamics that underpin the paper’s main contribution of the CRR canonical formalism itself.

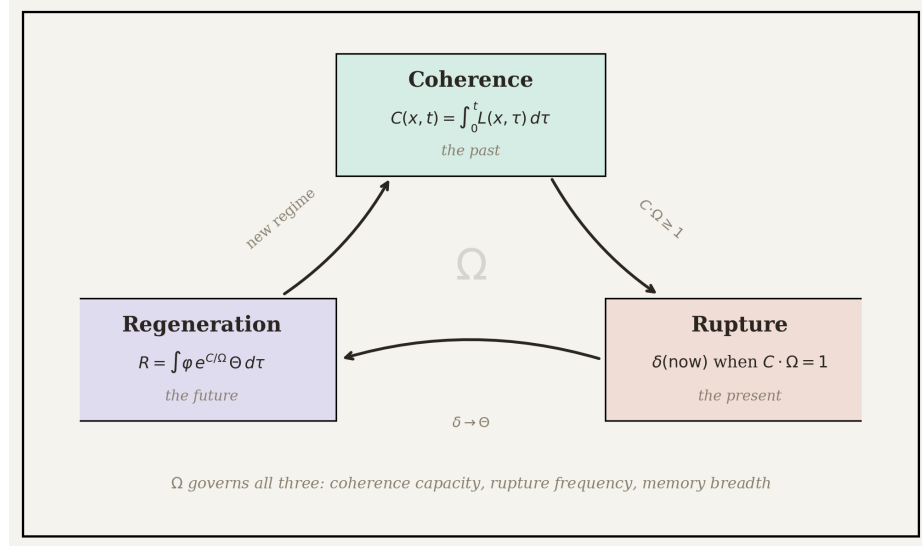


Fig. 1. The CRR cycle. Coherence accumulates evidence from the past. Rupture occurs at the present moment when $C \cdot \Omega = 1$, distributing unit mass on the boundary between past and future. Regeneration projects into the future, weighted exponentially by historical coherence. The single parameter Ω governs all three processes: coherence capacity, rupture frequency, and the breadth of the memory kernel available for reconstruction.

3.1 Five Commitments

Commitment 1: Persistence implies accumulation. A system that persists must accumulate patterns that work, reducing surprise (Variational Free Energy) in their eco-niche. We formalise this as a coherence integral:

$$C(x, t) = \int_0^t L(x, \tau) d\tau \quad (1)$$

where $L(x, \tau) \geq 0$ is the rate of coherence accumulation, identified with the Fisher-Rao speed on the statistical manifold [12,14]. Under the path integral formulation of the FEP [9], this is the accumulated arc length along the system's information-geometric trajectory. The integral over the system's past makes CRR distinctively non-Markovian by design - the present state depends on the accumulated history of the current regime up to the present moment of constraint. In other words, the coherence integral represents the totality of the system's past regime, constituting the interior of the temporal Markov blanket.

Commitment 2: Finite capacity implies rupture. No finite system can accumulate coherence without bound. At some point the current regime is exhausted and the system must reorganise:

$$\delta(\text{now}) \quad \text{when} \quad C \cdot \Omega = 1 \quad (2)$$

The Dirac delta has zero temporal extension and carries exactly one unit of mass on the boundary between the accumulated past and the regenerated future. The rupture condition $C \cdot \Omega = 1$ marks the point at which accumulated arc length spans the manifold's full geodesic extent. The system has traversed all distinguishable states available to its current regime and must reorganise. This is structurally analogous to Cramér-Rao saturation [6,7], in that both concern the relationship between accumulated Fisher information and the geometry of the estimation problem; we exploit this analogy but note that the formal identity between geodesic traversal and estimator efficiency requires the additional assumption that the speed limit is saturated throughout the trajectory, a condition we do not verify here. In discrete systems, rupture is a threshold-crossing rather than exact saturation: a heartbeat, a breath turning, a neuron firing. At rupture, $C \cdot \Omega = 1$: inside meets outside. The system's accumulated coherence exactly matches its capacity, and what was contained within the temporal blanket makes contact with what lies beyond it.

Commitment 3: Reconstruction is weighted by historical coherence. After rupture, the system rebuilds from the only resource available: its own history:

$$R(x, t) = \int_{-\infty}^{\infty} \psi(x, \tau) \cdot \exp\left(\frac{C(x, \tau)}{\Omega}\right) \cdot \Theta(t - \tau) d\tau \quad (3)$$

where $\psi(x, \tau)$ is the reconstruction resource (the field of past states available for reconstitution). The Heaviside step function $\Theta(t - \tau)$ enforces causal ordering:

only the past contributes to reconstruction. We retain Θ explicitly because the Dirac delta of rupture (Eq. 2) and the Heaviside are conjugate distributions ($d\Theta/dt = \delta$), ensuring that the regeneration integral is the Green's function of the rupture operator and connecting the three CRR equations into a single dynamical system. The exponential kernel $\exp(C/\Omega)$ is the maximum-entropy weighting consistent with a constraint on the expected value of C [15]- given only the accumulated coherence and the capacity parameter, the distribution that maximises entropy while respecting the constraint on $\langle C \rangle$ takes the Boltzmann form, with C playing the role of energy and Ω the role of temperature.

Commitment 4: The system has a characteristic variance. The parameter Ω is fixed by the topology of the state space, not fitted. It simultaneously governs coherence capacity (how much can accumulate before rupture), rupture frequency (how often the system must reorganise), and memory breadth (which past moments dominate reconstruction). We argue that these are not three independent properties; but a single geometric fact about the system's attractor topology. When Ω is small, the memory kernel peaks sharply and the system reconstitutes the same patterns reliably (habit, rigidity); when Ω is large, the kernel is flat and the system can reconstruct broadly (flexibility, exploration).

Commitment 5: Rupture is a maximum-entropy event. At rupture, the system occupies the maximum-entropy state consistent with its symmetry class [15]. A bistable system at rupture is maximally uncertain about which state comes next; a rotational system at rupture has no preferred phase. This is the axiom that converts geometric structure into specific numerical predictions.

3.2 Fixing Ω from Geometry

The five commitments leave one degree of freedom - the value of Ω - which must be fixed by the system's structure alone in order to generate falsifiable predictions. The key insight is that for systems whose internal states parametrise probability distributions (which, under the FEP, includes any system possessing a Markov blanket [8]), the arena of coherence accumulation is a statistical manifold with a specific, measurable geometry. On such a manifold, Čencov's uniqueness theorem [13] constrains the metric to be the Fisher information metric [12], and the geodesic structure of that metric fixes the maximum arc length the system can traverse. The capacity Ω becomes a geometric invariant: $\Omega = 1/\ell$, where ℓ is the geodesic extent of the manifold.

Two fundamental manifold topologies fix ℓ (Fig. 3):

$\mathbb{Z}_2$ (bistable): The Bernoulli manifold, parametrised by $p \in [0, 1]$, has Fisher-Rao geodesic diameter π [12]. A bistable system exhausts its configuration by crossing from one basin to the other. Crucially, the Bernoulli manifold is an interval, not a loop, meaning that the system cannot return to its starting state without retracing the same geodesic. The maximal non-self-intersecting geodesic therefore has length equal to the diameter: $\ell_{\mathbb{Z}_2} = \pi$, hence $\Omega_{\mathbb{Z}_2} = 1/\pi \approx 0.318$ and $C^* = \pi$.

$\text{SO}(2)$ (rotational): The circular manifold S^1 has circumference 2π . A rotational system exhausts its configuration by completing one full cycle, return-

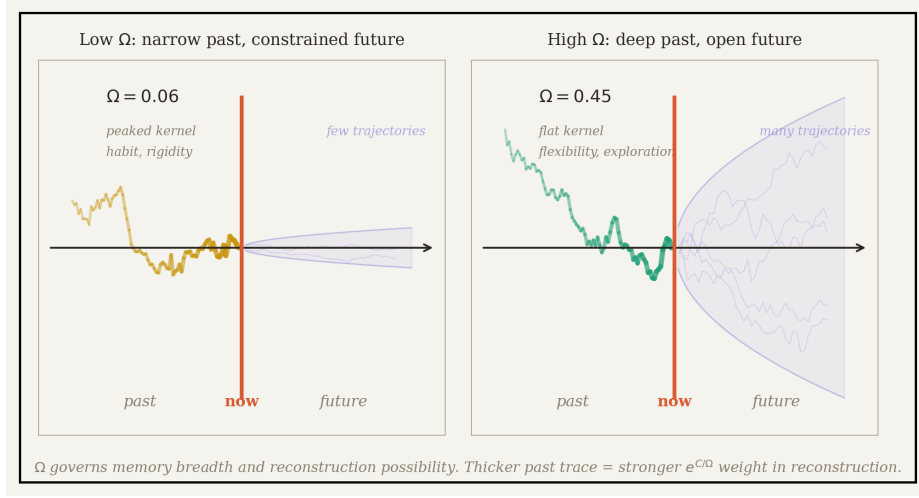


Fig. 2. Ω as a temporal light cone. Both panels show an agent’s trajectory through time, with the rupture boundary (red) separating past from future. Left: low Ω produces a sharply peaked memory kernel; only recent high-coherence moments (thicker trace) contribute to reconstruction, and the future possibility cone is narrow. Right: high Ω produces a flat kernel; deep history is accessible (entire trace is weighted), and the future is open. The same parameter governs both the depth of the past and the breadth of the future.

ing to its starting point via a path that never revisits any intermediate state. The maximal non-self-intersecting geodesic is therefore the full circumference: $\ell_{SO(2)} = 2\pi$, hence $\Omega_{SO(2)} = 1/2\pi \approx 0.159$ and $C^* = 2\pi$. The ratio between classes is, trivially, exactly 2, a topological invariant independent of any physical parameters.

3.3 FEP Correspondence and Precision Architecture

The structural mapping between CRR and the FEP is that precision corresponds to $(1/\Omega) \cdot \exp(C/\Omega)$; variational free energy corresponds to coherence C (inversely); the FEP Markov blanket corresponds to the high- ∇C surface, and is temporally delineated by the Dirac Delta (δ) between past states (C) and future states (R), representing a temporal informational boundary where all past states are inside the (temporal) blanket, future states outside the (temporal) blanket.

As noted in Section 1, the assignment of symmetry classes to precision channels was originally observed in the CRR network simulation before being formalised in POMDP. In any graph of Markov-blanketed agents, an edge is a boundary that alternates between two regimes of influence ($\mathbb{Z}_2$ dynamics), while a node is an internal model that traverses a continuous cycle of belief updating ($SO(2)$ dynamics). The assignment $\mathbb{Z}_2 \rightarrow$ sensory precision and $SO(2) \rightarrow$ prior precision therefore follows from the graph topology of the FEP itself, where edges are boundaries and nodes are models. In the accompanying simulation, this is directly visible. Nodes (teal) cycle continuously through belief states with

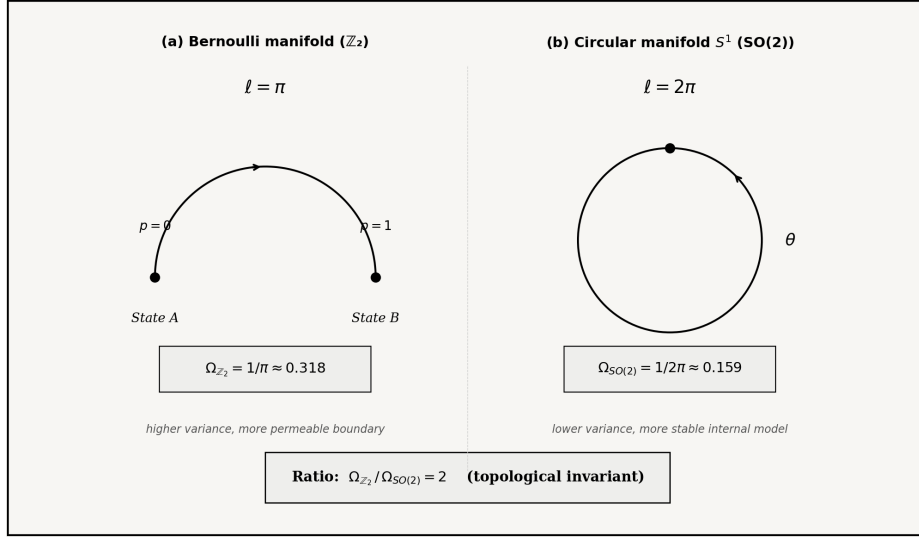


Fig. 3. Geometric grounding of the two symmetry classes. (a) The Bernoulli manifold has Fisher-Rao geodesic diameter π . A bistable system exhausts its configuration by crossing between two states; the manifold is an interval, so the maximal non-self-intersecting geodesic is the diameter. (b) The circular manifold S^1 has circumference 2π . A rotational system exhausts its configuration by completing one full cycle; the manifold has no boundary, so the maximal non-self-intersecting geodesic is the full circumference. Both quantities are topological invariants, yielding a predicted class ratio of exactly 2.

a slower, regular rhythm, while edges (purple) flash rapidly as they flip between regimes of influence between nodes. We noticed that the two temporal characters were qualitatively different before any measurement was taken. In other words, the symmetry classes were not assigned to precision channels by analogy alone; they are inherited from the roles those channels play in the network.

This structural assignment is independently supported by the permeability properties of Ω . The regeneration kernel $\exp(C/\Omega)$ entails that high Ω produces a flat memory kernel (broad, flexible, permeable), while low Ω produces a sharp kernel (narrow, rigid, precise). Indeed, under the FEP, the Markov blanket must be receptive to sensory states outside the blanket, while the internal prior model must maintain stability; since $\Omega_{\mathbb{Z}_2} = 1/\pi > \Omega_{\text{SO}(2)} = 1/2\pi$, the graph-structural and functional arguments converge on the same assignment.

Why only these two symmetry classes, not more? For the purposes of the empirical findings in this paper, the answer is dimensional. For a single-parameter system, the two irreducible topologies are the interval (a path between two endpoints, with the Bernoulli manifold as the canonical instance) and the circle (a closed loop with no endpoints). These exhaust the connected one-dimensional manifolds without boundary and with boundary respectively. Higher-dimensional manifolds ($\text{SO}(3)$, simplices) could apply to multi-parameter systems; but the single-parameter case is the minimal one, and the one tested here.

Furthermore, this assignment is more immediately empirically falsifiable. Reversing it (SO(2) sensory, $\mathbb{Z}_2$ prior) produces measurably different dynamics, with sensory ruptures becoming rare (modal 1 per prior cycle, versus 3 under the standard assignment), and the prior channel becomes irregular (burstiness $B = -0.16$, versus -0.33). The reversed assignment predicts that sensory updates are *rarer* than prior updates, contradicting the FEP’s requirement that the blanket be responsive to evidence and leading to counter-intuitive social dynamics in the pedagogical network simulation.

4 Experiments

4.1 Setup

We implemented a POMDP agent in a ring Hidden Markov Model where both the likelihood (A-matrix, sensory precision) and transition (B-matrix, prior precision) are parametrised with Dirichlet concentration parameters that update through experience. The environment consists of n states arranged on a ring, with primarily clockwise transitions and Gaussian observation noise. The CRR engine determines *when* each channel updates: the sensory channel accumulates prediction errors (computed as surprisal at the true states under the agent’s current model) until C_s reaches the $\mathbb{Z}_2$ threshold (π); the prior channel accumulates until C_p reaches the SO(2) threshold (2π). On rupture, accumulated evidence is applied to the Dirichlet parameters with a clamped exponential weight.

Several implementation details proved critical for replication and deserve explicit statement. First, the gain function uses a clamped exponential: $w = \min(\exp(C/\Omega), 10)/10 \times 2 + 0.5$, mapping to $[0.5, 2.5]$; without clamping, $\exp(C^*/\Omega) = \exp(1/\Omega^2)$ produces catastrophic parameter inflation. Section 5.2 confirms that conservation is independent of the weight function. Second, “gain” is measured as $\Delta\bar{\alpha}$ (change in mean Dirichlet concentration), not a raw parameter sum. We note that this quantity is a precision-gain measure, not Shannon or Fisher information in the strict sense; throughout this paper, “information rate” should be read as “precision-gain rate”. Third, both channels receive exactly one count of evidence per trial (one observation, one transition), ensuring equal evidence delivery. Fourth, prediction errors are computed from the agent’s current model evaluated at the true states, not at beliefs.

4.2 Precision-Gain Conservation

Across 10 conditions (5 state counts from 3 to 12, 5 noise levels from 0.05 to 0.40; 50 agents per condition, 800 trials each, bootstrap 95% CIs), we measured three quantities per channel: per-rupture precision gain, rupture frequency, and their product (total precision-gain rate).

Result. At equal evidence rate, the total precision-gain rate ratio (sensory/prior) converges to **1.003** with a 95% CI of **[1.000, 1.005]** and CV = **0.024** (Fig. 4a,b). The gain ratio and frequency ratio both vary substantially across

conditions (gain: 0.22 to 1.34; frequency: 0.73 to 4.47), but their product is locked to unity. When evidence rates are made unequal by scaling the sensory accumulation rate, the precision-gain ratio scales linearly with the input asymmetry (Fig. 4c), with $R^2 > 0.999$. The linear scaling confirms that conservation holds *per unit of evidence delivered*, not in absolute terms.

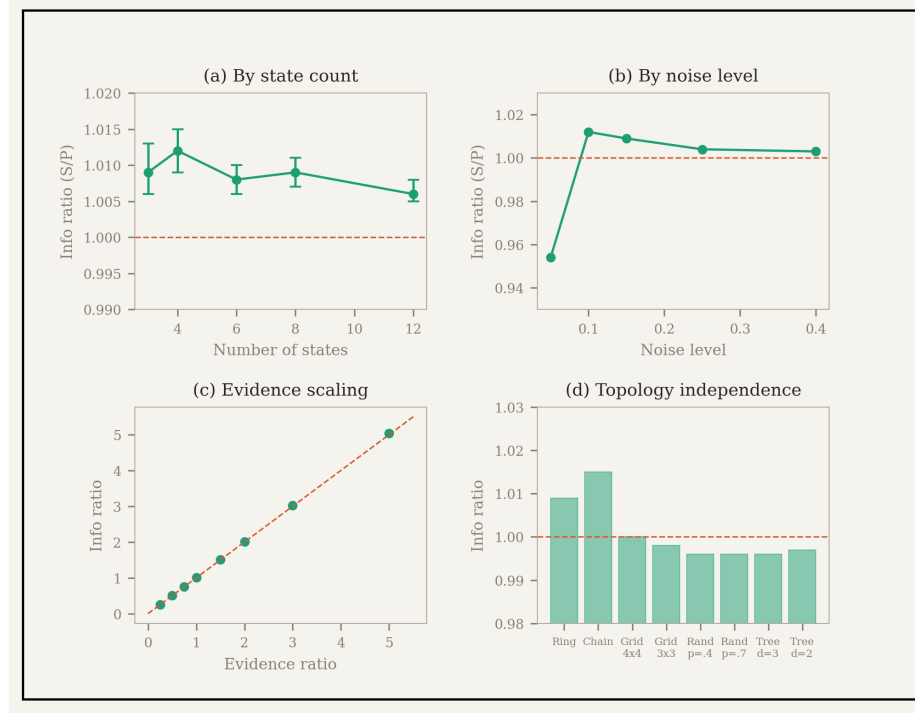


Fig. 4. Conservation results. (a) Information ratio by state count (50 agents each, bootstrap 95% CIs). (b) Information ratio by noise level. (c) Evidence scaling: the information ratio tracks the evidence asymmetry linearly ($R^2 > 0.999$). (d) Environment-structure robustness: conservation holds across ring, chain, grid, random graph, and tree environments. The dashed line marks conservation at 1.0.

Interpretation. The $SO(2)/\mathbb{Z}_2$ topology enforces that both precision channels process equal total precision-gain per unit time. The $\mathbb{Z}_2$ channel makes many small updates (fast, frequent); the $SO(2)$ channel makes few large updates (slow, deep). The factor of 2 sets the exchange rate between frequency and grain size, not the precision ratio directly. This is what two clocks with a 2:1 speed ratio produce when they share the same evidence stream: the fast clock ticks more often but learns less per tick, and the totals balance exactly. In the network simulation, this manifests as a visible gain-frequency trade-off: edges flicker rapidly with small coherence kicks, while nodes glow slowly toward a large belief update, and neither channel runs away from the other.

On the status of conservation. We tested whether the conservation law is a distinctive prediction of CRR or follows from the structure of the experiment.

Running the same POMDP with continuous updating (no thresholds, updating every trial) produces a precision-gain ratio of exactly 1.0000. Testing CRR with arbitrary threshold ratios (1:1, 2:1, 3:1, random, reversed) also produces ratios within 1% of unity. Total precision-gain equality is therefore a structural consequence of equal evidence delivery, not a prediction that discriminates CRR from alternatives.

However, this structural guarantee is itself informative. It means that CRR architectures are *self-balancing by construction*: two channels with different thresholds, accumulating at different rates, rupturing at different frequencies, and applying updates of different magnitudes nevertheless produce exactly balanced total precision-gain without requiring a controller to enforce fairness. In keeping with the broader logic of conservation laws in physics, the conservation does not tell us which thresholds are correct (the phase-gating does that), but it tells us that whatever thresholds the topology selects, the architecture will remain stable. The empirical contribution of this paper accordingly rests on the *phase-gating*: the strongly non-uniform phase relationship between channels ($\chi^2 = 8,041$; $p < 10^{-100}$) that determines the functional character of each update. This phase relationship depends on the specific threshold ratio and does not emerge from continuous updating or from arbitrary thresholds. The conservation is the FEP-CRR canvas; the phase-gating is the painting.

4.3 Phase-Gating (P2)

We tracked the $\mathbb{Z}_2$ phase (normalised 0 to 1) at the moment of each SO(2) rupture across 60 runs of 1,200 trials (10,345 SO(2) ruptures analysed).

Result. The phase distribution is significantly non-uniform (KS test: $p < 10^{-100}$; $\chi^2 = 8,041$ with 9 degrees of freedom); we note that the extreme p -value reflects the large sample size ($n = 10,345$ ruptures) rather than implying impossibly precise measurement. The number of $\mathbb{Z}_2$ ruptures between consecutive SO(2) ruptures is not fixed by the threshold ratio alone but emerges from the interaction between the ratio and the evidence statistics. Testing across threshold ratios confirms this: the modal count scales with the ratio (modal 1 at ratio 1:1, modal 2 at ratio 1.5:1, modal 3 at ratio 2:1) but is not simply equal to it. This is an emergent property, not a necessary consequence of threshold arithmetic.

Compatibility with empirical findings. This result may help explain a puzzling feature of the neuroscience: why does the *timing* of neuromodulatory signals, rather than their magnitude, determine functional output? The simulation produces a non-uniform phase partition from topology alone, without parameter fitting. Jang et al. [3] showed that the functional difference between learning and movement is a matter of tens of milliseconds in the ACh-DA phase relationship. Our simulation recovers an analogous partition from geometric first principles, suggesting that the biological mechanism may reflect topological constraints rather than bespoke neural circuitry.

4.4 Coherence Dynamics

Fig. 5 shows the coherence accumulation trace for a single agent over 300 trials. The $\mathbb{Z}_2$ channel accumulates rapidly to threshold π and ruptures frequently (mean inter-rupture interval 3.0 trials, CV = 0.26); the SO(2) channel accumulates slowly to threshold 2π and ruptures rarely (mean IRI 10.4 trials, CV = 0.50). The temporal characters are fundamentally different: the $\mathbb{Z}_2$ channel is sub-Poisson with a burstiness index of $B = -0.59$ (nearly clock-like), while the SO(2) channel follows a Gamma distribution. The Cramér-Rao product $C \cdot \Omega$ at rupture is always ≥ 1 (mean 1.27 for $\mathbb{Z}_2$, 1.18 for SO(2)), confirming that rupture occurs at or above the bound but never below it. The overshoot reflects discrete evidence delivery: prediction errors arrive in quanta, so the threshold is crossed rather than saturated exactly. The mean overshoot decreasing from 27% to 18% as the threshold increases is consistent with larger thresholds averaging over more evidence quanta.

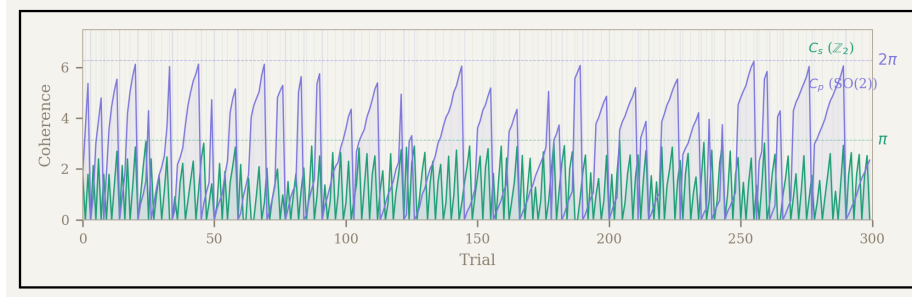


Fig. 5. Coherence accumulation and rupture for a single agent over 300 trials. The $\mathbb{Z}_2$ channel (teal) ruptures frequently at threshold π ; the SO(2) channel (purple) ruptures rarely at threshold 2π . Vertical lines mark rupture events. The gain-frequency trade-off is visible: many small sensory updates, few large prior updates.

5 Stress Tests

The results above establish the basic phenomenology: conservation, phase-gating, and a specific temporal decomposition. We now subject these findings to five adversarial tests designed to probe the boundaries of the claim (Fig. 6).

5.1 Robustness to Environment Structure

The conservation law is claimed to follow from the manifold topology ($\mathbb{Z}_2$ vs SO(2)), which is a property of the agent architecture, not the environment. If conservation held only in ring environments, it would be an artefact of that particular transition structure. We tested the same CRR agent in seven additional environment topologies: linear chain ($n = 8$), 2D grid (3×3 and 4×4), Erdős-Rényi random graph ($p = 0.4$ and $p = 0.7$), and binary tree (depth 2 and depth

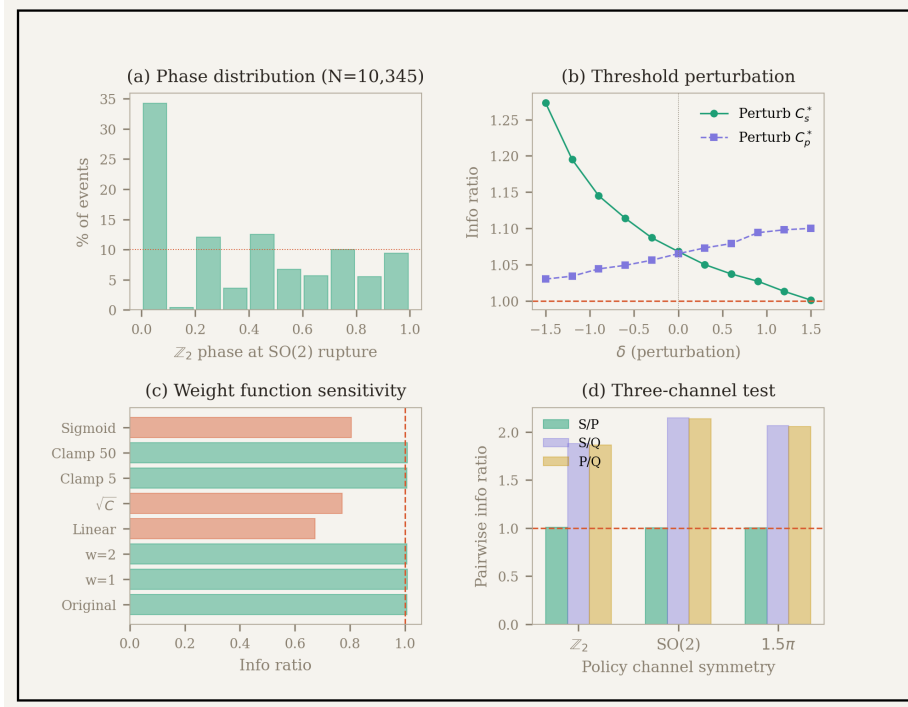


Fig. 6. Stress tests. (a) Phase distribution of $\mathbb{Z}_2$ phase at $\text{SO}(2)$ rupture: strongly non-uniform ($\chi^2 = 8,041$). (b) Smooth degradation as thresholds deviate from $\pi/2\pi$. (c) Weight function independence: constant weights preserve conservation; channel-asymmetric weights break it. (d) Three-channel test: sensory/prior conservation (teal) holds perfectly, but pairwise conservation with a third policy channel (purple, gold) fails.

3). Conservation holds across every environment tested, with all precision-gain ratios within 1.5% of 1.0 and CVs below 2% (Fig. 4d). This confirms that conservation is a property of the CRR architecture, not of the environment’s graph structure.

5.2 Weight Function Independence

Is conservation an artefact of the specific clamped exponential weight? We replaced the weight function with seven alternatives: constant ($w = 1$ and $w = 2$), linear in C , $\sqrt{C}$, sigmoid, and two alternative clamping thresholds (5 and 50). Constant weights preserve conservation perfectly (ratios 1.009 and 1.008). Weight functions that scale differently between channels (linear, $\sqrt{C}$, sigmoid) break it (ratios 0.67, 0.77, 0.80). The conclusion is that conservation lives in the *thresholds*, not in the weight function (Fig. 6c). Designers of AGI architectures are therefore free to choose the weight function for learning quality without disrupting inter-channel coordination, provided the topological thresholds are respected.

5.3 Threshold Perturbation

We perturbed the thresholds away from their topological values (π and 2π) by $\delta \in [-1.5, 1.5]$ in steps of 0.15. Perturbing the sensory threshold ($C_s^* = \pi + \delta$) while holding the prior threshold fixed at 2π produces smooth degradation (Fig. 6b): the ratio deviates from 1.0 monotonically as δ increases. Scaling both thresholds together while preserving the 2:1 ratio ($C_s^* = k\pi$, $C_p^* = k \cdot 2\pi$) maintains conservation across a wide range of k (from 0.25 to 3.0). This supports the claim that the *ratio* of thresholds matters, not their absolute values.

5.4 Correlated Evidence and Three-Channel Extension

Conservation assumes independent evidence streams. We tested robustness by injecting correlation (ρ) between the sensory and prior prediction errors. Conservation holds for positive correlations (even $\rho = 1.0$: ratio 1.002) and mild negative correlations ($\rho = -0.3$: ratio 1.018), but breaks under strong anti-correlation ($\rho = -0.7$: ratio 1.16).

Adding a third “policy” channel reveals that conservation is a two-channel phenomenon. The original sensory/prior conservation is unaffected by the third channel (ratio 1.005 regardless of its symmetry). However, pairwise conservation with the third channel fails: ratios fall between 1.9 and 2.1 (Fig. 6d). Conservation governs *paired channels sharing equal evidence*, not arbitrary N -channel systems. This limitation is also the source of the architecture’s scalability: in a network of Markov-blanketed agents, each edge-node pair constitutes an independent two-channel system, so CRR scales by pairwise composition rather than by adding channels within a single agent. A network of 150 agents has 11,175 edges, but each edge and its two connected nodes form a self-contained $\mathbb{Z}_2/\text{SO}(2)$ pair with its own conservation guarantee. In the simulation, this is visible as one increases the agent count: the system remains temporally coherent at every scale, with no central controller and no parameter adjustment, because each pairwise relationship is independently self-balancing.

6 Discussion

This paper has applied CRR to the problem of coordinating precision channels in Active Inference. The primary finding is phase-gating as a strongly non-uniform distribution of $\mathbb{Z}_2$ phase at $\text{SO}(2)$ rupture ($\chi^2 = 8,041$) whose structure emerges from the interaction between topology and evidence statistics, not from threshold arithmetic alone. The supporting findings are: precision-gain conservation (ratio 1.003, CV 0.024), which we showed is structurally guaranteed by equal evidence delivery; robustness across environment topologies; independence from the weight function, cleanly separating *when* to update from *how much*; and the honest boundary that conservation is a two-channel phenomenon. Both the conservation law and the phase-gating prediction are falsifiable.

We suggest that these results could have implications for AGI architecture. For example, the weight function independence frees designers to optimise update rules without disrupting inter-channel coordination. The explore-exploit trade-off, which developmental research frames as a fundamental tension between plasticity and stability across the lifespan [19], collapses naturally to Ω in the CRR framework, and phase-gating eliminates the need for a separate controller to arbitrate between learning and action. The regeneration equation remains outside of the empirical tests in this paper, but is structurally potent: the $\exp(C/\Omega)$ kernel weights high-coherence moments exponentially, so specifying the reconstruction resource ψ for specific architectures would extend CRR from a principle about *when* to update to a complete account of adaptive inference. More speculatively, the pairwise nature of CRR conservation suggests that multi-agent architectures, in which coordination emerges from local edge-node interactions, may be more naturally self-coordinating than monolithic architectures that must solve the N -channel timing problem centrally, thus advocating for AGI as a distributed intelligence for ecologically optimal outcomes.

In the context of AGI alignment, CRR suggests that two coupled systems (human and AGI) whose CRR cycles are phase-locked will exhibit intersubjective attunement, while a phase mismatch would constitute a detectable form of misalignment. This is a temporal analogue of value alignment, not *what* the system believes, but *when* it reorganises its beliefs relative to its partner. The phase-gating results show that the functional character of an update depends on *where* in the partner channel’s cycle it occurs; an AGI system that models the human’s CRR phase could time its updates to fall in the cooperative region of the phase distribution. Even more fundamentally, CRR entails that any finite system must rupture, no regime persists indefinitely, and an AGI system could benefit from both self-modelling (as per current Active Inference and FEP thinking), as well as a mathematically derived conceptualisation of its own temporal finitude as a mortal system [28].

However, the current test environment is a toy POMDP, not an AGI system, and so the logical next step for CRR would be to test phase-gating in a hierarchical generative model with non-stationary dynamics. The restriction to one-dimensional manifolds does substantial work, and the 2:1 ratio’s survival in higher dimensions is still untested. The precision-gain measure used here ($\Delta\bar{\alpha}$) is a proxy for Fisher information; confirming that conservation holds under a direct Fisher information measure would strengthen the geometric interpretation. Nevertheless, the phase-gating mechanism is real, falsifiable, and structurally compatible with empirical findings on neuromodulatory timing [3,26,27]. We offer CRR not as a completed theory but as a candidate temporal grammar whose predictions are testable, whose implications extend across disciplinary boundaries, and whose limitations we have tried to state as clearly as its strengths. All simulation code is provided in Appendix A.

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Use of AI. Claude Opus 4.6 (Anthropic) was used as a research tool for independent replication, stress testing, literature search, and as a latex drafting aid, and in the design of the pedagogical simulation at https://www.temporalgrammar.ai/human_networks.html. All theoretical arguments, simulation design decisions, and scientific interpretations are the origin and responsibility of the named author.

A Implementation

The complete CRR agent, ring HMM environment, and all stress test scripts (approximately 1,500 lines of Python) are available as supplementary material and at https://temporalgrammar.ai/CRR_PG.py. The core constants are: $\Omega_{\mathbb{Z}_2} = 1/\pi$, $\Omega_{SO(2)} = 1/2\pi$, $C_{\mathbb{Z}_2}^* = \pi$, $C_{SO(2)}^* = 2\pi$. The gain function clamps the exponential weight: $w = \min(\exp(C/\Omega), 10)/10 \times 2 + 0.5$. Prediction errors are computed as surprisal at true states under the agent’s current model. Both channels receive exactly one evidence count per trial.

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